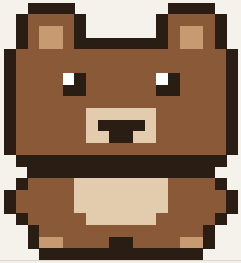
ROMA TRE · A.Y. 2026/27

Policy Evaluation Lab

Lecture 0: Course presentation and syllabus



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PART 1

The course



About the course

- **Policy Evaluation Lab** – Master's degree (LM), first semester
- Scientific area: ECON-02/A (Economic policy)
- 60 hours, 9 CFU, taught in English
- Timetable: 5.00–7.00pm, Room 20. **Lectures** on Monday and Thursday; **lab and discussion** (seminar) on Friday
- Course website (slides, datasets, exercises, announcements):
  - https://pel.carusomatteo.it
  - register with your university e-mail
- Official announcements are also posted on Moodle (available soon)

Why the need for a *policy*?

- In classical economics, State intervention is not needed: markets work by themselves
- This is rarely true
- We often need policies to:
  - ensure fairness and avoid inequality
  - reallocate and redistribute resources and endowments

  Markets alone cannot ensure this
- We also need policies to:
  - improve markets themselves (e.g. antitrust policies)
  - fix negative externalities (e.g. pollution)

> Policies to fix **market failures**

Why the *evaluation*?

- We can apply a policy, but then we want to understand:
  - whether it was the right policy
  - whether it could have been done better
  - how to improve it in the future
- Did it have the effect we were hoping for?
  - Did the health reform actually improve hospitals? Did people get better?
  - Did the minimum wage decrease employment? Or did it increase labour supply instead?
- We will never see what would have happened **without** the policy (the **counterfactual**), but we can make inference about it

> This is what **causal inference** is about

Why a *lab*?

- We want to use **real-world data** to see how to evaluate policies
- We will make use of software for econometric analysis
  - the choice will depend on your background: from Excel to R
- We will see together how researchers and academics use this software to evaluate policies, provide new evidence on social dynamics, and better understand society as a whole
- Each week, the **Friday** session is a lab and **class discussion** of a specific policy:
  - we will look at the data, discuss it, and think together about how it could have been done better

> **Seminars** and **discussions** in class

Learning outcomes

By the end of the course you will be able to:

1. **Frame** a **policy question** in terms of **potential outcomes** and **counterfactuals**
2. **Recognise selection bias** and explain why **correlation is not causation**
3. **Choose** the right **research design** for a given policy and data availability
4. **Estimate treatment effects** using several econometric methods (e.g. RCTs, matching, DiD, IV, RDD and synthetic control)
5. **Assess** the **identifying assumptions** and run the standard **validity checks**
6. **Read**, **replicate** and **critically discuss** an applied evaluation study

PART 2

Programme



Programme (I)

Week	Topic	Lab / case study
1	Causality and potential outcomes	Course presentation; software set-up
2	Selection bias and regression	OLS, controls; Lab 1: first steps in R
3	Randomised experiments	Bloom et al. (2013); Atkin et al. (2017); ITT, A/B tests
4	Selection on observables	Matching, propensity score, IPW
5	Difference-in-differences I	Card and Krueger (1994), minimum wage

Programme (II)

Week	Topic	Lab / case study
6	Difference-in-differences II	Event studies, staggered adoption
7	Instrumental variables	Angrist and Krueger (1991); LATE
8	Regression discontinuity	Sharp and fuzzy RDD, bandwidth choice
9	Synthetic control	Abadie et al. (2010), California Prop. 99
10	Wrap-up	Report presentations and exam review

The order may change slightly during the semester: check the website.

PART 3

Exam



Assessment

- **Report** on a policy intervention of your choice
  - either an empirical analysis in R with real-world data, or an argument based on explanation and evidence
  - the policy can be one you assess as **successful** or **unsuccessful**
  - either way, you have to argue your thesis properly, using **solid evidence**
- **Presentation** of your report in the last week of the course (week 10)
- **Oral exam**
  - written instead, if the number of students does not allow us to complete the orals on time

> Argue your thesis with **solid evidence**

PART 4

Materials



Reading list

Main textbooks

- Angrist J.D., Pischke J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.
- Cunningham S. (2021). *Causal Inference: The Mixtape*. Yale University Press. Free online: https://mixtape.scunning.com

Further reading

- Angrist J.D., Krueger A.B. (1999). "Empirical strategies in labor economics". *Handbook of Labor Economics*, Vol. 3A, ch. 23.
- OECD (2025). *Implementation Toolkit for the OECD Recommendation on Public Policy Evaluation*. OECD Publishing, Paris. Link (PDF)

Software

- Labs are run in **R** (open source, so you can work on it yourselves) and **Excel**
  - R + RStudio: free, https://posit.co/download/rstudio-desktop
- **Stata** will be used only for a few examples
- Bring a laptop to the lab sessions
- Interactive simulations of each method: https://carusomatteo.it/econvis.html

The course website: registration

1. Go to https://pel.carusomatteo.it
2. Register with your **university e-mail**, student ID (*matricola*), degree programme and a password of your choice
3. Enter the **course access code** given in class
4. From then on, log in with your e-mail and password

ACCESS CODE

Written on the board in class

The course website: a tour

- **Home**: announcements, timetable, location and office hours
- **Materials**: everything, week by week
  - slides are posted **before** each lecture; datasets and code **after**
  - some material (e.g. solutions) is released at a later date
- **Exercises**: problem sets for each method and interactive practice on EconVis
- **Exam**: assessment rules and exam sessions
- **Resources**: textbooks, software guides and datasets
- **Account**: your details and password

> Check the **announcements** regularly

PART 5

Let’s start



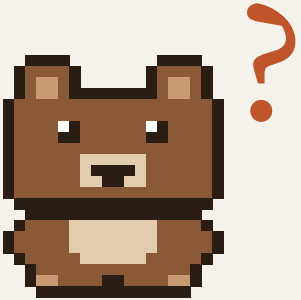
Getting to know you

A short, **anonymous** questionnaire (about two minutes):

- what you studied before
- which areas of economics interest you
- how confident you feel about maths and statistics
- which maths, statistics or econometrics courses you took
- your professional goals after the degree
- any additional comments

https://pel.carusomatteo.it/survey.php

Scan the QR code:
we will look at the results
together



Questions?

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